Shubham Farera



Senior Software Engineer (SDE-3) • Distributed Systems & Backend Architect • Java, Spring Boot & Kafka Specialist

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EXECUTIVE PROFESSIONAL SUMMARY

Senior Software Engineer with 8.5+ years of experience architecting, designing, and delivering high-throughput, fault-tolerant distributed systems and enterprise microservices. Deep expertise across High-Level Design (HLD) and Low-Level Design (LLD), applying SOLID principles, GoF design patterns, and clean architecture to scale production systems to 10M+ daily transactions with 99.99% uptime.

Proven specialist in distributed systems engineering: architecting event-driven pipelines with Apache Kafka, implementing distributed transactions via the Saga Pattern (Orchestration & Choreography) and Outbox Pattern, engineering low-latency search with Elasticsearch, and scaling databases via sharding and master-slave replication. Adept in navigating CAP and PACELC trade-offs to deliver resilient, sub-150ms latency platforms across Healthcare, FinTech, and Logistics.

TECHNICAL COMPETENCIES & ARCHITECTURE

- **</> Languages & Core :** **Java (8 / 11 / 17)**, Data Structures & Algorithms (DSA), Object-Oriented Analysis & Design (OOAD), Multithreading & Concurrency.
- **Distributed Systems :** **Apache Kafka**, Saga Pattern (Orchestration & Choreography), Transactional Outbox, **Elasticsearch**, Redis Caching, Consistent Hashing, CAP & PACELC Theorems, ZooKeeper.
- **Architecture & LLD :** **Microservices Architecture**, High-Level Design (HLD), Low-Level Design (LLD), **SOLID Principles**, GoF Design Patterns (Strategy, Factory, Observer, Decorator, State), Clean Architecture.
- **Frameworks & DB :** **Spring Boot**, Spring Cloud, Hibernate / JPA, RESTful APIs, **MySQL**, **Oracle**, PostgreSQL, Database Sharding & Master-Slave Replication, SQL Optimization.
- **Cloud & DevOps :** **Microsoft Azure (AZ-900 Certified)**, Docker, CI/CD (Jenkins Pipelines), CodeQL Static Analysis, SonarQube Quality Gates, Git & GitHub, JUnit, Mockito, Agile / Scrum.

PROFESSIONAL EXPERIENCE

IT Analyst (Senior Backend Engineer) @ Tata Consultancy Services

Oct 2023 – Present

- Architected and deployed scalable, event-driven microservices using Java 17, Spring Boot, and Apache Kafka, processing 10M+ daily events for healthcare claims and policy workflows with 99.99% availability.
- Implemented distributed transaction workflows across billing and policy underwriting services using the Saga Pattern (Orchestration & Choreography) and Outbox Pattern, guaranteeing eventual consistency across heterogeneous datastores without 2PC overhead.
- Engineered low-latency search and retrieval pipelines using Elasticsearch, designing custom analyzer mappings and typeahead query DSLs that reduced p99 search latency by 65% (from 450ms to <150ms).
- Refactored legacy monolithic services into modular, extensible microservices adhering to SOLID principles and GoF design patterns (Strategy, Factory, Observer, Decorator), increasing unit test coverage from 45% to 88% using JUnit and Mockito.
- Scaled database architecture by implementing MySQL/Oracle Master-Slave read replication and horizontal table partitioning, paired with multi-tier Redis distributed caching, reducing database read contention by 55%.
- Engineered system resilience and fault tolerance utilizing Resilience4j circuit breakers, token-bucket rate limiting, and graceful degradation, applying CAP and PACELC trade-offs to withstand high-volume traffic surges.
- Streamlined CI/CD deployment automation using Jenkins pipelines, enforcing automated CodeQL static analysis and SonarQube quality gates to maintain 0 critical/high vulnerabilities in production.

PROFESSIONAL EXPERIENCE (CONTINUED)

Application Development Senior Analyst @ Accenture
Sept 2021 – Aug 2023

- Singapore-based Asset Rebalancing and Investment System: Developed backend microservices for automated portfolio rebalancing workflows, executing real-time asset allocation calculations for $500M+ in assets under management (AUM).
- Implemented secure, idempotent RESTful APIs integrating with third-party banking gateways and financial settlement networks, reducing API transaction failure rates by 40%.
- Optimized complex SQL queries, transaction management, and Hibernate ORM configurations for high-volume transactions, improving end-of-day batch settlement execution time by 35%.
- Enforced zero-defect code quality standards and led vulnerability remediation across 12 services using SonarQube, maintaining an 'A' grade code reliability rating.

Application Development Analyst @ Accenture
May 2020 – Aug 2021

- Australia-based Logistics Order Management System: Designed and developed core order processing and shipment tracking REST APIs using Java, Spring Boot, and JPA, handling 200K+ daily orders.
- Applied State and Strategy design patterns to manage multi-stage order lifecycles and shipment state transitions, eliminating concurrency race conditions during status updates.
- Enhanced database indexing, query execution plans, and connection pool configurations, reducing average API response times by 30% during peak delivery windows.

Associate Software Engineer @ Accenture
Feb 2018 – Apr 2020

- US-based Healthcare and Child Support Project: Developed core Java modules, data transformation utilities, and scheduled batch jobs supporting enterprise case-management workflows.
- Authored unit and integration test suites using JUnit and Mockito, ensuring 85%+ test coverage across delivered components within Agile sprints.
- Gained deep hands-on expertise in OOP, data structures, multithreading, and Git version control, consistently resolving client-reported defect tickets ahead of SLA deadlines.

KEY ARCHITECTURAL PLATFORMS & DISTRIBUTED SYSTEMS

Distributed Event-Driven Insurance Platform
High-throughput Spring Boot microservices backed by Apache Kafka and the Saga pattern, handling 10M+ daily events with 99.99% availability.

Elasticsearch Search & Indexing Engine
Low-latency policy and member search with custom analyzers and typeahead lookup, slashing query response times by 65%.

Asset Rebalancing & Investment Platform
Real-time portfolio rebalancing engine executing automated allocations for $500M+ in assets under management with idempotent guarantees.

Logistics Order Lifecycle Engine
High-throughput order tracking and event handling pipeline managing 200K+ daily shipments with zero race conditions.

EDUCATION

Bachelor of Engineering – Information Technology • Shri Govindram Seksaria Institute of Technology and Science (SGSITS Indore, MP) — 2013 – 2017 | **CGPA: 7.1**

Key Coursework: Data Structures & Algorithms, Object-Oriented Analysis & Design, Database Management Systems (DBMS), Operating Systems, Computer Networks, Software Engineering.

CERTIFICATIONS & LEADERSHIP DISTINCTIONS

Microsoft Certified: Azure Fundamentals (AZ-900) • Cloud Concepts, Architecture, Governance & Compliance **[Certified]**

- **National-Level Sports Representation:** Represented SGSITS Indore at the National-Level Table Tennis Tournament held at BITS Pilani K.K Birla Goa Campus.
- **Competitive Leadership:** Appointed Captain of state-level CBSE Football Cluster Team (2011); competed in 3 consecutive state-level cluster tournaments (2009–2011).
- **Academic Distinction:** Achieved All India Rank **370 (AIR)** in the National Science Olympiad (NSO).